CE 310 — Week 6 Pre-Lab Quiz

Week: 6 | Topic: Empirical Percentiles Across CE Practice Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers LARGE/SMALL/PERCENTILE, exceedance probability, duration curves, and how ACI 318, the 85th-percentile speed rule, and AACE contingency practice apply these tools. Review the Week 6 Background reading before completing this quiz.

Questions

Q1. For a dataset of n = 500 concrete strength records, =SMALL(D2:D501, 50) returns:

(A) The average strength of the weakest 50 records
(B) The 50th weakest individual record — the empirical 10th percentile
(C) The 50th percentile (median) strength
(D) The strength value exceeded by 50% of all records

Q2. The exceedance probability $P(X > x)$ is defined as:

(A) The probability that X equals exactly x
(B) The fraction of records less than x
(C) The fraction of records that exceed x
(D) The number of standard deviations between x and the mean

Q3. On a duration curve built by ranking a variable from highest to lowest, the x-axis represents:

(A) Time, in chronological order
(B) The variable's own value
(C) Rank — 1 for the highest value, n for the lowest
(D) Cumulative probability from 0 to 1

Q4. ACI 318's concrete compliance rule (no more than 1-in-10 tests below spec) is fundamentally:

(A) A Normal-distribution assumption about concrete strength

(B) A 10th-percentile compliance test, applied empirically to real test data
(C) A requirement that the mean strength exceed the spec by 10%
(D) Only valid when the sample size exceeds 1,000

Q5. Compared to the standard deviation, the interquartile range (IQR) is:

(A) More sensitive to outliers, since it uses the extreme values
(B) Robust to outliers, since it depends only on the middle 50% of the data
(C) Only defined for Normally distributed data
(D) Mathematically identical to the standard deviation for any dataset

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Q6. Which Excel formula correctly computes the percentage of records in F2:F501 (CostPerCY) that exceed $250?

(A) =COUNTIF(F2:F501,">250")/500*100
(B) =COUNTIF(F2:F501,">250")
(C) =COUNTIF(F2:F501,">250")*100
(D) =SUM(F2:F501>250)/500

Q7. =SUMPRODUCT((F2:F501-250)*(F2:F501>250)) computes:

(A) The total number of records above $250
(B) The average cost of records above $250
(C) The total dollar amount by which all above-threshold records exceed $250, summed
(D) The percentage of records above $250

Q8. Quantile regression differs from ordinary least squares (OLS) regression in that it:

(A) Cannot be used with more than one predictor variable
(B) Estimates a specified conditional quantile (e.g., the 90th percentile) instead of the conditional mean
(C) Requires the response variable to be Normally distributed
(D) Is mathematically identical to OLS, just renamed

Q9. A colleague computes the 10th percentile of concrete strength for Supplier Alpha using =SMALL(D2:D501,24) and gets 4,181 psi, then computes it again using =PERCENTILE(FILTER(D2:D501,B2:B501="Alpha"),0.10) and gets 4,186 psi. Why do the two answers differ slightly?

(A) One of the two formulas must contain an error

(B) SMALL returns an exact observed data point at a whole rank; PERCENTILE interpolates between two adjacent sorted values — both are legitimate, they just answer slightly different versions of the question
(C) PERCENTILE only works on the full dataset, not a filtered subset
(D) The two functions require different units

Q10. The empirical P(CostPerCY > $250) is 16.80%, while Week 5's lognormal model predicted 15.8%. What is the most defensible interpretation?

(A) The empirical result must be wrong, since the model was fit carefully
(B) The two are close enough that the lognormal model remains a reasonable approximation, though the empirical value is the one to trust for an exact decision
(C) A 1.0 percentage-point gap proves the lognormal model is completely invalid
(D) The gap means the sample size (500) was too small to compute either value reliably

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